CS4221/5421: Tutorial 6 — Functional Dependencies

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AY25/26 S1

Identifying Functional Dependencies

Example Relation, $R = \{A, B, C\}$ and $\Sigma = \{\{A\} \rightarrow \{B\}, \{B\} \rightarrow \{C\}\}$

The column in red decides the color of the row: The rows having same values in the red column, have same color.

Task: Check which other columns are following "same value if same color" for each deciding (red) column. The columns for which this holds, are implied by the red column.

A	B	C
a1	b1	c1
a1	b1	c1
a2	b2	c2
a2	b2	c2
a3	b3	c2
a3	b3	c2
a4	b1	c1
a4	b1	c1
a5	b2	c2

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$\{A\} \rightarrow \{A, B, C\}$

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a2	b2	c2
a3	b3	c2
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a4	b1	c1
a4	b1	c1
a5	b2	c2

$\{B\} \rightarrow \{B, C\}$

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$\{A\} \rightarrow \{A, B, C\}$

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a2	b2	c2
a2	b2	c2
a3	b3	c2
a3	b3	c2
a4	b1	c1
a4	b1	c1
a5	b2	c2

$\{B\} \rightarrow \{B, C\}$

A	B	C
a1	b1	c1
a1	b1	c1
a2	b2	c2
a2	b2	c2
a3	b3	c2
a3	b3	c2
a4	b1	c1
a4	b1	c1
a5	b2	c2

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a3	b3	c2
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a4	b1	c1
a5	b2	c2

$\{A\} \rightarrow \{A, B, C\}$

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a1	b1	c1
a2	b2	c2
a2	b2	c2
a3	b3	c2
a3	b3	c2
a4	b1	c1
a4	b1	c1
a5	b2	c2

$\{B\} \rightarrow \{B, C\}$

A	B	C
a1	b1	c1
a1	b1	c1
a2	b2	c2
a2	b2	c2
a3	b3	c2
a3	b3	c2
a4	b1	c1
a4	b1	c1
a5	b2	c2

$\{C\} \rightarrow \{C\}$

Rules for Functional Dependencies (Armstrong's Axioms)

W, X, Y, Z mentioned here are sets of attributes.

Sound & complete inference system

- **Reflexivity**: If $Y \subseteq X$, then $X \to Y$.
- **Augmentation**: If $X \to Y$, then $X \cup Z \to Y \cup Z$ for any Z.
- **Transitivity**: If $X \to Y$ and $Y \to Z$, then $X \to Z$.

Common derived rules (from the three above)

- **Union / Additivity**: If $X \to Y$ and $X \to Z$, then $X \to Y \cup Z$.
- **Decomposition / Projectivity**: If $X \to Y \cup Z$, then $X \to Y$ and $X \to Z$.
- **Pseudotransitivity**: If $X \to Y$ and $W \cup Y \to Z$, then $W \cup X \to Z$.
- **Composition**: If $X \to Y$ and $Z \to W$, then $X \cup Z \to Y \cup W$.
- **Self-determination**: $X \to X$.

How we use them here

- *LHS reduction*: test if $X \setminus \{A\} \to Y$ holds via closure using the axioms.
- *Redundancy removal*: test if an FD $X \to Y$ is implied by the rest (i.e., $Y \subseteq X^+$ computed from the others).

Question 1.a.

Consider the following schema R and the set of functional dependencies Σ.

$$R = \{A, B, C, D, E\},$$

$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}.$

Question 1.a.

Consider the following schema R and the set of functional dependencies Σ.

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Show that $\Sigma \vdash \{C, D\} \rightarrow \{A, B, C, D, E\}$ using Armstrong's axioms.

1. $\{C, D\} \rightarrow \{A, B\}$
2. $\{A, B, C\} \rightarrow \{E\}$
3. $\{B, C, D\} \rightarrow \{A, E\}$
4. $\{B, D\} \rightarrow \{A\}$
5. $\{A, C\} \rightarrow \{B, E\}$

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3. $\{B, C, D\} \rightarrow \{A, E\}$
4. $\{B, D\} \rightarrow \{A\}$
5. $\{A, C\} \rightarrow \{B, E\}$
6. $\{C, D\} \rightarrow \{A, B, C, D\}$ (Augmentation of (1) with $\{C, D\}$)

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$$R = \{A, B, C, D, E\},$$

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2. $\{A, B, C\} \rightarrow \{E\}$
3. $\{B, C, D\} \rightarrow \{A, E\}$
4. $\{B, D\} \rightarrow \{A\}$
5. $\{A, C\} \rightarrow \{B, E\}$
6. $\{C, D\} \rightarrow \{A, B, C, D\}$ (Augmentation of (1) with $\{C, D\}$)
7. $\{B, C, D\} \rightarrow \{A, B, C, D, E\}$ (Augmentation of (3) with $\{B, C, D\}$)

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Consider the following schema R and the set of functional dependencies Σ.

$$R = \{A, B, C, D, E\},$$

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Show that $\Sigma \vdash \{C, D\} \rightarrow \{A, B, C, D, E\}$ using Armstrong's axioms.

1. $\{C, D\} \rightarrow \{A, B\}$
2. $\{A, B, C\} \rightarrow \{E\}$
3. $\{B, C, D\} \rightarrow \{A, E\}$
4. $\{B, D\} \rightarrow \{A\}$
5. $\{A, C\} \rightarrow \{B, E\}$
6. $\{C, D\} \rightarrow \{A, B, C, D\}$ (Augmentation of (1) with $\{C, D\}$)
7. $\{B, C, D\} \rightarrow \{A, B, C, D, E\}$ (Augmentation of (3) with $\{B, C, D\}$)
8. $\{A, B, C, D\} \rightarrow \{B, C, D\}$ (Reflexivity)

Question 1.a.

Consider the following schema R and the set of functional dependencies Σ.

$$R = \{A, B, C, D, E\},$$

$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}$.
Show that $\Sigma \vdash \{C, D\} \rightarrow \{A, B, C, D, E\}$ using Armstrong's axioms.

1. $\{C, D\} \rightarrow \{A, B\}$
2. $\{A, B, C\} \rightarrow \{E\}$
3. $\{B, C, D\} \rightarrow \{A, E\}$
4. $\{B, D\} \rightarrow \{A\}$
5. $\{A, C\} \rightarrow \{B, E\}$
6. $\{C, D\} \rightarrow \{A, B, C, D\}$ (Augmentation of (1) with $\{C, D\}$)
7. $\{B, C, D\} \rightarrow \{A, B, C, D, E\}$ (Augmentation of (3) with $\{B, C, D\}$)
8. $\{A, B, C, D\} \rightarrow \{B, C, D\}$ (Reflexivity)
9. $\{C, D\} \rightarrow \{B, C, D\}$ (Transitivity between (6) and (8))

Question 1.a.

Consider the following schema R and the set of functional dependencies Σ.

$$R = \{A, B, C, D, E\},$$

$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}$.
Show that $\Sigma \vdash \{C, D\} \rightarrow \{A, B, C, D, E\}$ using Armstrong's axioms.

1. $\{C, D\} \rightarrow \{A, B\}$
2. $\{A, B, C\} \rightarrow \{E\}$
3. $\{B, C, D\} \rightarrow \{A, E\}$
4. $\{B, D\} \rightarrow \{A\}$
5. $\{A, C\} \rightarrow \{B, E\}$
6. $\{C, D\} \rightarrow \{A, B, C, D\}$ (Augmentation of (1) with $\{C, D\}$)
7. $\{B, C, D\} \rightarrow \{A, B, C, D, E\}$ (Augmentation of (3) with $\{B, C, D\}$)
8. $\{A, B, C, D\} \rightarrow \{B, C, D\}$ (Reflexivity)
9. $\{C, D\} \rightarrow \{B, C, D\}$ (Transitivity between (6) and (8))
10. $\{C, D\} \rightarrow \{A, B, C, D, E\}$ (Transitivity between (9) and (7))

Question 1.b.

$$R = \{A, B, C, D, E\},$$

$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}$.
Show that $\Sigma \models \{C, D\} \rightarrow \{A, B, E\}$ using **The Chase**.

Question 1.b.

$$R = \{A, B, C, D, E\},$$

$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}$.
Show that $\Sigma \models \{C, D\} \rightarrow \{A, B, E\}$ using **The Chase**.

A	B	C	D	E
α	α	α	α	α
a_1	b_1	c_1	d_1	e_1

Setup

Question 1.b.

$$R = \{A, B, C, D, E\},$$

$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}$.
Show that $\Sigma \models \{C, D\} \rightarrow \{A, B, E\}$ using **The Chase**.

A	B	C	D	E
α	α	α	α	α
a_1	b_1	c_1	d_1	e_1

Setup

A	B	C	D	E
α	α	α	α	α
a_1	b_1	α	α	e_1

Initialization

Question 1.b.

$$R = \{A, B, C, D, E\},$$

$\Sigma = \{\{C, D\} \to \{A, B\}, \{A, B, C\} \to \{E\}, \{B, C, D\} \to \{A, E\}, \{B, D\} \to \{A\}, \{A, C\} \to \{B, E\}\}$.
Show that $\Sigma \models \{C, D\} \to \{A, B, E\}$ using **The Chase**.

A	B	C	D	E
α	α	α	α	α
a_1	b_1	c_1	d_1	e_1

Setup

A	B	C	D	E
α	α	α	α	α
a_1	b_1	α	α	e_1

Initialization

A	B	C	D	E
α	α	α	α	α
α	α	α	α	e_1

$\{C, D\} \to \{A, B\}$

Question 1.b.

$$R = \{A, B, C, D, E\},$$

$\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}$.
Show that $\Sigma \models \{C, D\} \rightarrow \{A, B, E\}$ using **The Chase**.

A	B	C	D	E
α	α	α	α	α
a_1	b_1	c_1	d_1	e_1

Setup

A	B	C	D	E
α	α	α	α	α
a_1	b_1	α	α	e_1

Initialization

A	B	C	D	E
α	α	α	α	α
α	α	α	α	e_1

$\{C, D\} \rightarrow \{A, B\}$

A	B	C	D	E
α	α	α	α	α
α	α	α	α	α

$\{A, C, D\} \rightarrow \{E\}$

We stop here as we have already reached $\{C, D\} \rightarrow \{A, B, E\}$ (the values in the second row to look at).

Question 1.c.

Show that $\{A, C\} \rightarrow \{D\}$ does not hold on R with Σ using Algorithm 1.

Algorithm 1: Attribute Closure Algorithm

Input: S, Σ
Output: S^+
$\Omega \leftarrow \Sigma$; `// unused`
$\Gamma \leftarrow S$; `// closure`
while *there exists* $(X \rightarrow Y) \in \Omega$ *such that* $X \subseteq \Gamma$ **do**
    $\Omega \leftarrow \Omega \setminus \{X \rightarrow Y\}$;
    $\Gamma \leftarrow \Gamma \cup Y$;
return Γ;

Using Algorithm 1 we get $\{A, C\}^+ = \{A, B, C, E\}$. As $D \notin \{A, C\}^+$, $\{A, C\} \nrightarrow \{D\}$ From Σ we get $\{A, C\} \rightarrow \{B, E\}$. $\{B\}$ determines nothing other than $\{B\}$. Same holds for $\{E\}$. Also, we see $\{B, E\}$ determines nothing other than $\{B, E\}$.

Question 1.d.

Find all keys of R w.r.t. Σ.

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Find all keys of R w.r.t. Σ. If we go algorithmically, we need to find the closures of all subsets of R.

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$\{A\}^+ = \{A\}$

$\{B\}^+ = \{B\}$

$\{C\}^+ = \{C\}$

$\{D\}^+ = \{D\}$

$\{E\}^+ = \{E\}$

$\{A, B\}^+ = \{A, B\}$

$\{A, C\}^+ = \{A, B, C, E\}$

$\{A, D\}^+ = \{A, D\}$

$\{A, E\}^+ = \{A, E\}$

$\{B, C\}^+ = \{B, C\}$

$\{B, D\}^+ = \{A, B, D\}$

$\{B, E\}^+ = \{B, E\}$

$\{C, D\}^+ = \{A, B, C, D, E\}$

$\{C, E\}^+ = \{C, E\}$

$\{D, E\}^+ = \{D, E\}$

$\{A, B, C\}^+ = \{A, B, C, E\}$

$\{A, B, D\}^+ = \{A, B, D\}$

$\{A, B, E\}^+ = \{A, B, E\}$

$\{A, C, D\}^+ = \{A, B, C, D, E\}$

$\{A, C, E\}^+ = \{A, B, C, E\}$

$\{A, D, E\}^+ = \{A, D, E\}$

$\{B, C, D\}^+ = \{A, B, C, D, E\}$

$\{B, C, E\}^+ = \{B, C, E\}$

$\{B, D, E\}^+ = \{A, B, D, E\}$

$\{C, D, E\}^+ = \{A, B, C, D, E\}$

$\{A, B, C, D\}^+ = \{A, B, C, D, E\}$

$\{A, B, C, E\}^+ = \{A, B, C, E\}$

$\{A, B, D, E\}^+ = \{A, B, D, E\}$

$\{A, C, D, E\}^+ = \{A, B, C, D, E\}$

$\{B, C, D, E\}^+ = \{A, B, C, D, E\}$

$\{A, B, C, D, E\}^+ = \{A, B, C, D, E\}$

Question 1.e.

Find one minimal cover of $\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}$

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Step 1: Simplify the RHS

1. $\{C, D\} \rightarrow \{A\}$
2. $\{C, D\} \rightarrow \{B\}$
3. $\{A, B, C\} \rightarrow \{E\}$
4. $\{B, C, D\} \rightarrow \{A\}$
5. $\{B, C, D\} \rightarrow \{E\}$
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6. $\{B, D\} \rightarrow \{A\}$
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8. $\{A, C\} \rightarrow \{E\}$

Step 2: Simplify the LHS

9. $\{C, D\} \rightarrow \{A\}$
10. $\{C, D\} \rightarrow \{B\}$
11. $\{A, C\} \rightarrow \{E\}$ From (8)
12. $\{B, D\} \rightarrow \{A\}$ From (6)
13. $\{C, D\} \rightarrow \{E\}$ From $\{C, D\}^+$
14. $\{B, D\} \rightarrow \{A\}$
15. $\{A, C\} \rightarrow \{B\}$
16. $\{A, C\} \rightarrow \{E\}$

Question 1.e.

Find one minimal cover of $\Sigma = \{\{C, D\} \rightarrow \{A, B\}, \{A, B, C\} \rightarrow \{E\}, \{B, C, D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B, E\}\}$

Step 1: Simplify the RHS

1. $\{C, D\} \rightarrow \{A\}$
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10. $\{C, D\} \rightarrow \{B\}$
11. $\{A, C\} \rightarrow \{E\}$ From (8)
12. $\{B, D\} \rightarrow \{A\}$ From (6)
13. $\{C, D\} \rightarrow \{E\}$ From $\{C, D\}^+$
14. $\{B, D\} \rightarrow \{A\}$
15. $\{A, C\} \rightarrow \{B\}$
16. $\{A, C\} \rightarrow \{E\}$

Remove redundancies

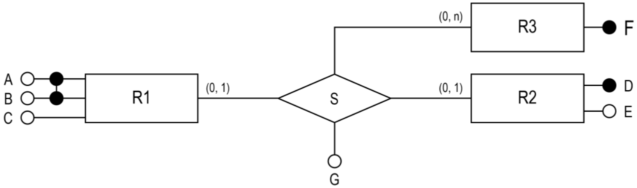
17. $\cancel{\{C, D\} \rightarrow \{A\}}$ As it can be derived from (10) and (14)
18. $\{C, D\} \rightarrow \{B\}$
19. $\cancel{\{A, C\} \rightarrow \{E\}}$ As we have it in (16)
20. $\cancel{\{B, D\} \rightarrow \{A\}}$ As we have it in (14)
21. $\cancel{\{C, D\} \rightarrow \{E\}}$ (10) and (14) $\implies$ $\{C, D\} \rightarrow \{A, C\}$. Now transitivity with (16).
22. $\{B, D\} \rightarrow \{A\}$
23. $\{A, C\} \rightarrow \{B\}$
24. $\{A, C\} \rightarrow \{E\}$

So we get one minimal cover:

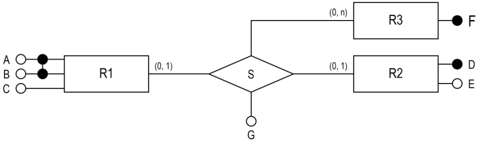
$$\{\{C, D\} \rightarrow \{B\}, \{B, D\} \rightarrow \{A\}, \{A, C\} \rightarrow \{B\}, \{A, C\} \rightarrow \{E\}\}$$

Q 2.

Consider the following entity-relationship diagram. Find all functional dependencies that hold on R.



Q 2. Solution



- From $R1$ we have $\{A, B\} \to C$.
- From $R2$ we have $\{D\} \to \{E\}$.
- From $R3$ we do not have anything extra.
- From participation of $R1$ and $R2$ in S, $\{A, B\} \to \{D\}$ and also $\{D\} \to \{A, B\}$. We also have $\{A, B\} \to \{G\}$ and $\{D\} \to \{G\}$.
- From participation of $R1$, $R2$ and $R3$ in S, we have $\{A, B\} \to \{F\}$, and $\{D\} \to \{F\}$.

Codes

You can take a look at this repository that will help you create random FD problems for your practice, and also given a csv file, you can find functional dependencies:
https://github.com/pratik2358/fucntional_dep

Questions?

Drop a mail at: pratik.karmakar@u.nus.edu